

TalentWise Advisory

Employee Attrition Risk Prediction

What is the business about?

TalentWise Advisory is a mid-sized consulting and business services firm with teams in sales, customer support, operations, finance, marketing, IT, and human resources. The company delivers project-based services to corporate clients and depends heavily on skilled employees, client knowledge, and consistent team performance.

What problem is the business trying to solve?

The main business problem is that managers often recognize employee attrition risk too late. By the time an employee resigns, the company has fewer options to address workload concerns, improve support, or offer career development opportunities.

What decision can machine learning help the business make?

Machine learning can help TalentWise predict whether an employee is likely to leave the company. The model will provide an early warning signal so that managers can review the situation and decide whether a supportive action is appropriate.

What is the target variable in the dataset?

The target variable is Left_Company. A value of 1 means the employee left the company, while a value of 0 means the employee stayed.

What are the input features?

The input features are Department, Role_Level, Employment_Type, Tenure_Months, Monthly_Salary, Overtime_Hours_Last_Month, Training_Hours_Last_Quarter, Performance_Rating, Engagement_Score, Manager_Support_Score, Remote_Work_Days_Per_Week, Absence_Days_Last_Quarter, Promotion_Last_Year, Commute_Distance_KM, Workload_Score and Salary_Satisfaction.

Why is this prediction useful for the business?

This prediction is useful because the business can use it to prioritize retention conversations, identify workload problems, and allocate HR resources more effectively.

Business Interpretation of Model Results

How well did the Decision Tree model perform?

The Decision Tree model showed moderate performance in predicting employee attrition. Based on the evaluation results, the model achieved approximately 79.17% accuracy, 44.44% precision, 28.57% recall, and 34.78% F1-score for employees who left the company.

Because the business goal is to identify employees at risk of leaving before resignation occurs, the low recall for the attrition class is an important limitation. The model missed many employees who actually left the company, which may reduce the company's ability to provide early support or intervention.

Is there a large difference between training accuracy and testing accuracy?

No, there is not a large difference between the training accuracy and testing accuracy. The controlled Decision Tree model achieved approximately 86.46% training accuracy and 79.17% testing accuracy, with a difference of about 7.29% between the two results. This relatively small gap suggests that the model was able to generalize reasonably well to unseen employee data rather than simply memorizing the training data.

However, the evaluation metrics show that the model still had difficulty identifying employees who actually left the company. Although the overall accuracy was relatively high, the recall for the attrition class was only about 28.57%, meaning many actual attrition cases were missed. This indicates that while the model generalized reasonably well overall, its predictive performance for detecting employee attrition still has room for improvement.

Does the model show signs of overfitting? Why or why not?

The model shows only mild signs of overfitting because the training accuracy is higher than the testing accuracy. The controlled Decision Tree achieved approximately 86.46% training accuracy and 79.17% testing accuracy, which means the model performed slightly better on the employee data it learned from compared to unseen employee data.

However, the difference between the two accuracies is relatively small at about 7.29%, suggesting that the model was still able to generalize reasonably well rather than simply memorizing the training data.

Despite this, the evaluation metrics show that the model still struggled to identify employees who actually left the company. The recall for the attrition class was only about 28.57%, meaning many actual attrition cases were missed. This suggests that although severe overfitting was reduced, the model's predictive performance for employee attrition still has room for improvement.

Which evaluation metric is most important for this business problem?

For this employee attrition problem, recall is the most important evaluation metric because the company wants to identify as many employees who may leave the company as possible.

Recall measures how many employees who actually left were correctly identified by the model. This is important because if the model fails to detect employees who are planning to leave, the company may lose the chance to provide support or take action early.

What do false positives mean in this business context?

A false positive means the model predicts that an employee will leave the company, but the employee actually stays. In this situation, HR managers may incorrectly assume that the employee is at risk of leaving and spend unnecessary

time or resources on retention efforts such as coaching sessions, workload adjustments, engagement discussions, or career development plans.

Too many false positives may reduce operational efficiency because the company may focus retention efforts on employees who are not actually planning to leave.

What do false negatives mean in this business context?

A false negative means the model predicts that an employee will stay with the company, but the employee actually leaves. In this situation, the company may fail to recognize that the employee was at risk of leaving and lose the opportunity to provide support or intervention before the resignation happens.

This can lead to:

- recruitment and training costs
- project disruption
- loss of experienced employees and organizational knowledge
- reduced productivity and team morale

Too many false negatives may reduce the company's ability to proactively manage employee retention and workforce planning.

Which features were most important in the Decision Tree model?

Some of the most important features in the Decision Tree model are Manager_Support_Score, Engagement_Score, Department_Sales, Workload_Score, Tenure, Salary, Training_Hours and Commute_Distance. These features had higher importance scores, meaning they played a bigger role in helping the model predict whether an employee will leave or stayed.

How can these important features help the business make better decisions?

These important features can help TalentWise Advisory better understand which employees may be at higher risk of leaving and why. For example, low engagement scores, low manager support, high overtime hours, heavy workloads, lack of promotion opportunities, or low salary satisfaction may indicate a higher attrition risk.

By using these insights, HR managers and department leaders can make better business decisions, such as:

- reviewing employee workloads
- improving manager support and communication
- offering career development or training opportunities
- increasing employee engagement initiatives
- prioritizing retention efforts for employees who may be at higher risk of leaving

These actions can help reduce employee turnover, improve employee satisfaction, support workforce stability, and reduce recruitment and training costs.

What is one possible limitation or bias in the model or dataset?

One possible limitation is that the dataset is synthetic and may not fully represent real-world employee behavior or workplace situations. Important real-life factors such as personal career goals, family responsibilities, workplace culture, economic conditions, or external job opportunities are not included in the dataset.

In addition, some features such as engagement scores or manager support scores are based on surveys and may contain response bias because employees may not always answer honestly or consistently. The model may also learn historical workplace patterns that could unintentionally create bias toward certain departments, job roles, or employee groups.

Why should human judgment still be used when making business decisions based on model results?

Human judgment should still be used because the Decision Tree model showed only moderate performance and still produced several incorrect predictions. The model may incorrectly predict that an employee will leave when they actually stay, or fail to identify employees who are truly at risk of leaving.

Because of these prediction errors, relying only on the model could lead to poor HR decisions, unnecessary interventions, or missed opportunities to support employees who need help.

HR managers and business leaders can use their experience, communication skills, and understanding of workplace situations to review the model's predictions and consider other important factors that are not included in the dataset, such as employee motivation, personal circumstances, workplace culture, or external job opportunities.

Therefore, the model should be used as a decision-support tool to assist managers rather than replace human decision-making.